Measure Theoretic Probability
Lesson 11
The Borel–Cantelli Lemmas

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Outline

1. Setting and Statements
2. Borel–Cantelli I
3. Two Tools for Borel–Cantelli II
4. Borel–Cantelli II
5. Examples and Remarks
6. Summary and Exercises

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Setting and Interpretation

Setting

Throughout, $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $A_1, A_2, \cdots \in \mathcal{F}$. Think of A_n as "something happens in round n of a game" (for example, you win round n).

The event "infinitely often"

$$\{A_n \text{ i.o.}\} = \limsup_n A_n = \bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m.$$

Notation warning

Although written with an n, the event $\{A_n \text{ i.o.}\}$ does **not** depend on n: it is a statement about the **whole sequence**. Its complement is "only finitely many A_n occur":

$$\{A_n \text{ i.o.}\}^c = \{A_n \text{ f.o.}\} = \liminf_n A_n^c.$$

The Two Lemmas

Borel–Cantelli I

If $A_1, A_2, \cdots \in \mathcal{F}$ satisfy

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty,$$

then

$$\mathbb{P}\big(A_n \text{ i.o.}\big) = \mathbb{P}\Big(\limsup_n A_n\Big) = 0.$$

Equivalently, **finitely many** of the A_n occur with probability 1.

Borel–Cantelli II

If $A_1, A_2, \cdots \in \mathcal{F}$ are **independent** and

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty,$$

then

$$\mathbb{P}\big(A_n \text{ i.o.}\big) = \mathbb{P}\Big(\limsup_n A_n\Big) = 1.$$

So **infinitely many** of the A_n occur with probability 1.

Remark

BC I needs no independence. BC II needs both divergence of the series **and** independence.

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Motivating Example: A Scary-Looking Game

The game

In round n you bet 2^n units. With probability $\dfrac{2^n}{2^n+1}$ you win and net $+1$; with probability $\dfrac{1}{2^n+1}$ you lose your stake 2^n.

Is it fair?

Expected gain in round n:

$$1 \cdot \frac{2^n}{2^n+1} - 2^n \cdot \frac{1}{2^n+1} = 0.$$

The win probability tends to 1, yet there is **always** a positive chance of losing everything in each round.

Question

Let $A_n = \{\text{you lose in round } n\}$, so $\mathbb{P}(A_n) = \dfrac{1}{2^n+1}$. Will you lose infinitely often?

Remark

The game is closely related to a martingale (a topic for later), but it is not one.

Applying Borel–Cantelli I to the Game

The series converges

Shrinking the denominator enlarges each term, so

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \sum_{n=1}^{\infty} \frac{1}{2^n + 1} \;\leq\; \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n = \frac{1/2}{1 - 1/2} = 1 < \infty.$$

Conclusion

By BC I, $\mathbb{P}(A_n \text{ i.o.}) = 0$: with probability 1 you lose only **finitely many** times. Eventually you win $+1$ every round, and total winnings tend to $+\infty$.

The catch

You must be able to stay in the game and keep staking ever larger amounts. The lemma describes the probability space, not your bankroll.

Proof of Borel–Cantelli I

Proof

Assume $\sum_n \mathbb{P}(A_n) < \infty$. Then the tail sums $\sum_{k \geq n} \mathbb{P}(A_k) \to 0$ as $n \to \infty$. For **every** $n \geq 1$,

$$\limsup_m A_m = \bigcap_{N \geq 1} \bigcup_{k \geq N} A_k \subseteq \bigcup_{k=n}^{\infty} A_k.$$

By monotonicity and then countable subadditivity (the A_k need not be disjoint),

$$\mathbb{P}\Big(\limsup_m A_m\Big) \leq \mathbb{P}\Big(\bigcup_{k=n}^{\infty} A_k\Big) \leq \sum_{k=n}^{\infty} \mathbb{P}(A_k).$$

The left side is a fixed number that does not depend on n. Letting $n \to \infty$ the right side tends to 0, so $\mathbb{P}(\limsup_m A_m) \leq 0$. Since probabilities are non-negative,

$$\mathbb{P}\Big(\limsup_m A_m\Big) = 0. \qquad \square$$

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Tool 1: Independence Passes to Complements

Lemma

If A and B are independent, then A^c and B^c are independent.

Proof

By De Morgan and the inclusion–exclusion formula,

$$\mathbb{P}(A^c \cap B^c) = 1 - \mathbb{P}(A \cup B) = 1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(A \cap B).$$

Using independence $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$,

$$\mathbb{P}(A^c \cap B^c) = 1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(A)\mathbb{P}(B) = \big(1 - \mathbb{P}(A)\big)\big(1 - \mathbb{P}(B)\big) = \mathbb{P}(A^c)\mathbb{P}(B^c). \quad \square$$

What we really need

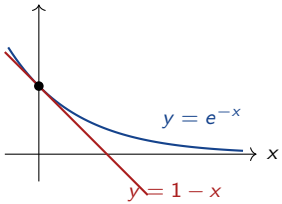
BC II uses this for **finitely many** events at a time: if $A_n, \ldots, A_k$ are independent, so are $A_n^c, \ldots, A_k^c$. This follows from Lesson 10: apply the main theorem to the π-systems $\{A_i\}$, whose generated σ-fields $\{\varnothing, A_i, A_i^c, \Omega\}$ contain A_i^c.

Tool 2: An Elementary Inequality

Lemma

For every real x (in particular for all $x \geq 0$),

$$1 - x \;\leq\; e^{-x}.$$



Proof

Let $f(x) = e^{-x}$. Then $f(0) = 1$ and $f'(0) = -1$, so the tangent line to f at $(0, 1)$ is $y = 1 - x$. Since $f''(x) = e^{-x} > 0$, f is convex, and a convex function lies above each of its tangent lines. □

Remark

The Taylor series $1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots$ suggests the inequality, but showing the remainder is non-negative is awkward. The tangent-line argument is cleaner.

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Proof of Borel–Cantelli II (1/2): Reduction

Setup

Let $A_1, A_2, \dots$ be independent with $\sum_n \mathbb{P}(A_n) = \infty$, and put $A = \limsup_n A_n$. It suffices to show $\mathbb{P}(A^c) = 0$. By De Morgan,

$$A^c = \bigcup_{n=1}^{\infty} \bigcap_{m=n}^{\infty} A_m^c.$$

Strategy

By countable subadditivity,

$$\mathbb{P}(A^c) \;\le\; \sum_{n=1}^{\infty} \mathbb{P}\Big(\bigcap_{m=n}^{\infty} A_m^c \Big),$$

so it is enough to show that **each** term $\mathbb{P}\big(\bigcap_{m \ge n} A_m^c\big)$ is 0.

The difficulty

Independence only lets us factor **finite** intersections. So we first truncate the infinite intersection at a finite k using monotonicity.

Proof of Borel–Cantelli II (2/2): The Estimate

Fix n and let $k \geq n$

The intersection shrinks as more sets are added, so

$$\mathbb{P}\Big(\bigcap_{m=n}^{\infty} A_m^c\Big) \;\leq\; \mathbb{P}\Big(\bigcap_{m=n}^{k} A_m^c\Big) = \prod_{m=n}^{k} \mathbb{P}(A_m^c) = \prod_{m=n}^{k} (1 - \mathbb{P}(A_m)) \;\leq\; \prod_{m=n}^{k} e^{-\mathbb{P}(A_m)} = \exp\Big(-\sum_{m=n}^{k} \mathbb{P}(A_m)\Big).$$

(Tool 1 gives the product; Tool 2 gives the last inequality.)

Let $k \to \infty$

The left side does not depend on k. Since $\sum_{m=1}^{\infty} \mathbb{P}(A_m) = \infty$ and removing the finitely many terms $m < n$ (each in $[0, 1]$) cannot make a divergent series of non-negative terms converge, $\sum_{m=n}^{k} \mathbb{P}(A_m) \to \infty$. Hence the right side tends to 0:

$$\mathbb{P}\Big(\bigcap_{m=n}^{\infty} A_m^c\Big) = 0 \qquad \text{for every } n.$$

Conclusion

$\mathbb{P}(A^c) \leq \sum_n 0 = 0$, so $\mathbb{P}(\limsup_n A_n) = 1$. □

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Example 1: Heads Infinitely Often

Setting

Flip a p-coin independently forever, with $0 < p < 1$. Let $A_n = \{\text{flip } n \text{ is heads}\}$.

Check the hypotheses

- The A_n are independent (independent flips).
- $\sum_n \mathbb{P}(A_n) = \sum_n p = \infty$, since we add the constant $p > 0$ forever.

Conclusion (BC II)

$\mathbb{P}(A_n \text{ i.o.}) = 1$: heads occurs infinitely often with probability 1.

Example 2: Runs of Ten Million Heads

First attempt fails

Let $A_n = \{\text{flips } n, \ldots, n+N-1 \text{ are all heads}\}$ with $N = 10^7$. These windows **overlap**, so the A_n are **not** independent and BC II does not apply.

Fix: use non-overlapping blocks

Split the flips into consecutive disjoint blocks of length N: block n consists of flips $(n-1)N+1, \ldots, nN$. Let

$$A_n = \{\text{block } n \text{ is all heads}\}.$$

Check the hypotheses

- The A_n depend on disjoint sets of flips, hence are independent.
- $\mathbb{P}(A_n) = p^N = p^{10^7}$: tiny, but a positive constant, so $\sum_n \mathbb{P}(A_n) = \infty$.

Conclusion (BC II)

$\mathbb{P}(A_n \text{ i.o.}) = 1$: runs of ten million heads occur infinitely often, with probability 1. We threw away information (overlapping runs) and still got the result.

Why Independence Is Needed in BC II

A counterexample without independence

Let $A \in \mathcal{F}$ with $\mathbb{P}(A) = \frac{1}{2}$ and set $A_n = A$ for all n. Then

$$\sum_n \mathbb{P}(A_n) = \sum_n \tfrac{1}{2} = \infty,$$

but $\limsup_n A_n = A$, so $\mathbb{P}(\limsup_n A_n) = \frac{1}{2} \neq 1$.

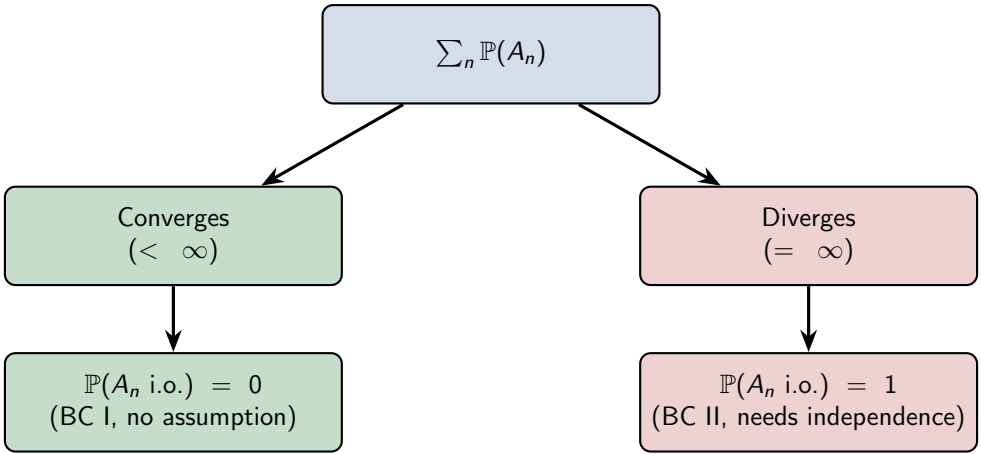
The converse of BC I is false in general

Divergence of $\sum_n \mathbb{P}(A_n)$ alone does **not** give $\mathbb{P}(A_n \text{ i.o.}) > 0$ in the strong sense, as the example shows. Independence is what turns divergence into probability one.

A zero–one dichotomy

For **independent** events, the two lemmas together give: $\mathbb{P}(A_n \text{ i.o.}) \in \{0, 1\}$, and it equals 0 if $\sum_n \mathbb{P}(A_n) < \infty$ and 1 if $\sum_n \mathbb{P}(A_n) = \infty$.

The Two Lemmas Side by Side



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Summary

Key results

Concept / Result	Content
$\{A_n \text{ i.o.}\}$	$\limsup_n A_n = \bigcap_n \bigcup_{m \geq n} A_m$; a property of the whole sequence
Borel–Cantelli I	$\sum \mathbb{P}(A_n) < \infty \Rightarrow \mathbb{P}(A_n \text{ i.o.}) = 0$ (no independence needed)
Borel–Cantelli II	independent and $\sum \mathbb{P}(A_n) = \infty \Rightarrow \mathbb{P}(A_n \text{ i.o.}) = 1$
Proof of BC I	monotonicity, countable subadditivity, tail sums $\to 0$
Proof of BC II	complements, $1 - x \leq e^{-x}$, truncate at k, let $k \to \infty$
Applications	game with stakes 2^n; heads i.o.; runs of 10^7 heads via disjoint blocks

Exercises

1. Let $\mathbb{P}(A_n) = 1/n^2$. What does BC I say about $\{A_n \text{ i.o.}\}$? Now let A_n be independent with $\mathbb{P}(A_n) = 1/n$. What does BC II say?
2. Prove that if $A_1, \ldots, A_n$ are independent, then $A_1^c, \ldots, A_n^c$ are independent (use the main theorem of Lesson 10).
3. Show that BC I still holds for an arbitrary measure μ with $\sum_n \mu(A_n) < \infty$ (that is, $\mu(\limsup_n A_n) = 0$). Does the proof use $\mu(\Omega) = 1$?
4. In the game of this lesson, let B_n be the event that you win round n. Compute $\sum_n \mathbb{P}(B_n)$ and discuss why BC II cannot be applied directly to B_n unless independence of rounds is assumed.
5. For independent fair coin flips, show that the pattern HTH appears infinitely often with probability 1.

Outlook

Next lesson

Tail events: for a sequence of events or σ-fields, look at what happens "from some point out to infinity", intersect over all starting points, and study the resulting **tail σ-field**.

Goal: Kolmogorov's zero–one law

The event $\{A_n \text{ i.o.}\}$ is an example of a tail event. For independent sequences, Kolmogorov's 0–1 law says every tail event has probability 0 or 1, and BC I and II are a first glimpse of this phenomenon.